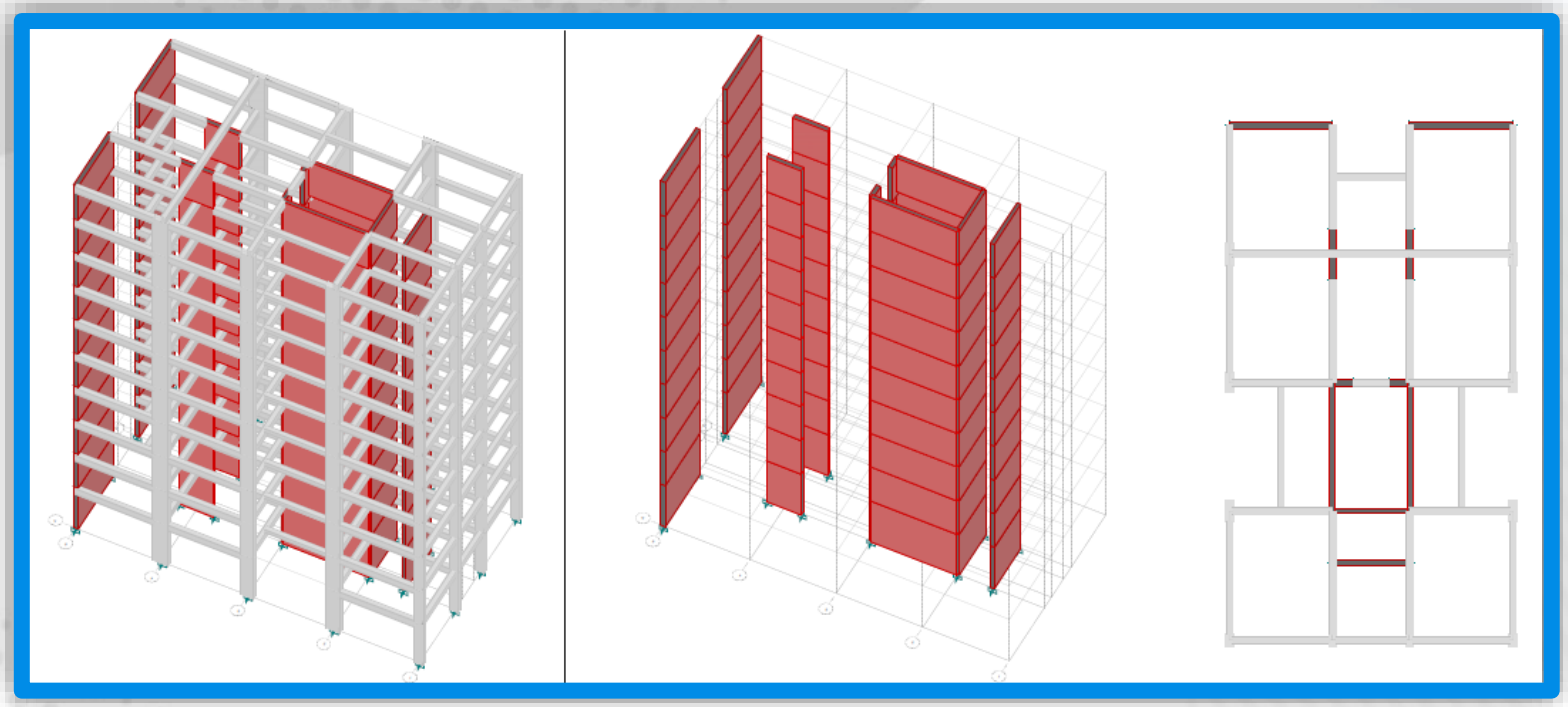


ANÁLISIS Y DISEÑO DE EDIFICACIONES DE CONCRETO ARMADO CON ETABS

SESIÓN 2 – COMBINACIONES DE DISEÑO

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COMBINACIONES DE DISEÑO

5.3—Load factors and combinations

5.3.1 Required strength U shall be at least equal to the effects of factored loads in Table 5.3.1, with exceptions and additions in 5.3.3 through 5.3.13.

Table 5.3.1—Load combinations

Load combination	Equation	Primary load
$U = 1.4D$	(5.3.1a)	D
$U = 1.2D + 1.6L + 0.5(L_r \text{ or } S \text{ or } R)$	(5.3.1b)	L
$U = 1.2D + 1.6(L_r \text{ or } S \text{ or } R) + (1.0L \text{ or } 0.5W)$	(5.3.1c)	L_r or S or R
$U = 1.2D + 1.0W + 1.0L + 0.5(L_r \text{ or } S \text{ or } R)$	(5.3.1d)	W
$U = 1.2D + 1.0E + 1.0L + 0.2S$	(5.3.1e)	E
$U = 0.9D + 1.0W$	(5.3.1f)	W
$U = 0.9D + 1.0E$	(5.3.1g)	E

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2.4 LOAD COMBINATIONS FOR ALLOWABLE STRESS DESIGN

2.4.1 Basic Combinations. Loads listed herein shall be considered to act in the following combinations; whichever produces the most unfavorable effect in the building, foundation, or structural member shall be considered. Effects of one or more loads not acting shall be considered. Seismic load effects shall be combined with other loads in accordance with Section 2.4.5. Wind and seismic loads need not be considered to act simultaneously. Refer to Sections 1.4, 2.4.5, 12.4, and 12.14.3 for the specific definition of the earthquake load effect E.

Increases in allowable stress shall not be used with the loads or load combinations given in this standard unless it can be demonstrated that such an increase is justified by structural behavior caused by rate or duration of load.

1. D
2. $D + L$
3. $D + (L_r \text{ or } S \text{ or } R)$
4. $D + 0.75L + 0.75(L_r \text{ or } S \text{ or } R)$
5. $D + (0.6W)$
6. $D + 0.75L + 0.75(0.6W) + 0.75(L_r \text{ or } S \text{ or } R)$
7. $0.6D + 0.6W$

ASCE 7

2.3 LOAD COMBINATIONS FOR STRENGTH DESIGN

2.3.1 Basic Combinations. Structures, components, and foundations shall be designed so that their design strength equals or exceeds the effects of the factored loads in the following combinations. Effects of one or more loads not acting shall be considered. Seismic load effects shall be combined loads in accordance with Section 2.3.6. Wind and seismic loads need not be considered to act simultaneously. Refer to Sections 1.4, 2.3.6, 12.4, and 12.14.3 for the specific definition of the earthquake load effect E. Each relevant strength limit state shall be investigated.

1. $1.4D$
2. $1.2D + 1.6L + 0.5(L_r \text{ or } S \text{ or } R)$
3. $1.2D + 1.6(L_r \text{ or } S \text{ or } R) + (L \text{ or } 0.5W)$
4. $1.2D + 1.0W + L + 0.5(L_r \text{ or } S \text{ or } R)$
5. $0.9D + 1.0W$
6. $1.2D + E_v + E_h + L + 0.2S$
7. $0.9D - E_v + E_h$
6. $1.2D + E_v + E_{mh} + L + 0.2S$
7. $0.9D - E_v + E_{mh}$

ASCE 7

Where the prescribed seismic load effect, $E = f(E_v, E_h)$ (defined in Section 12.4.2 or 12.14.3.1) is combined with the effects of other loads, the following seismic load combinations shall be used:

6. $1.2D + E_v + E_h + L + 0.2S$
7. $0.9D - E_v + E_h$

Where the seismic load effect with overstrength, $E_m = f(E_v, E_{mh})$, defined in Section 12.4.3, is combined with the effects of other loads, the following seismic load combination for structures shall be used:

6. $1.2D + E_v + E_{mh} + L + 0.2S$
7. $0.9D - E_v + E_{mh}$

❑ Considerando sobreresistencia

Combination 5: $(1.2 + 0.2S_{DS})D + \Omega_o Q_E + 1.0L + 0.2S$, and

Combination 7: $(0.9 - 0.2S_{DS})D + \Omega_o Q_E$.

COMBINACIONES DE DISEÑO

- **Efecto del Sismo**

$$E = E_h + E_v \qquad (12.4\text{-}1)$$

$$E = E_h - E_v \qquad (12.4\text{-}1)$$

- **Efecto del sismo Horizontal**

$$E_h = \rho Q_E \qquad (12.43)$$

❑ ρ = Factor de redundancia

- **Efecto del sismo Horizontal**

❑ **Se considera**

$$E_v = 0.2 S_{DS} \cdot D \qquad (12.4\text{-}4a)$$

❑ **Para el resto de casos**

$$E_v = 0.3 S_{av} \cdot D \qquad (12.4\text{-}4b)$$

EXCEPTIONS:

1. Where the option to incorporate the effects of vertical seismic ground motions using the provisions of Section 11.9 is required elsewhere in this standard, the vertical seismic load effect, E_v, shall be determined in accordance with Eq. (12.4-4b) as follows:

$$E_v = 0.3 S_{av} D \qquad (12.4\text{-}4b)$$

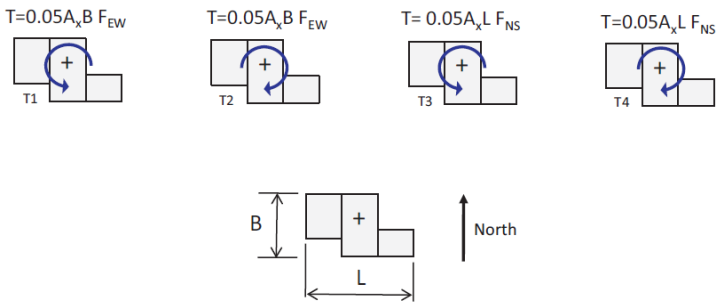
where

S_{av} = design vertical response spectral acceleration obtained from Section 11.9.3, and
D = effect of dead load.

2. The vertical seismic load effect, E_v, is permitted to be taken as zero for either of the following conditions:
   a. In Eqs. (12.4-1), (12.4-2), (12.4-5), and (12.4-6) for structures assigned to Seismic Design Category B.
   b. In Eq. (12.4-2) where determining demands on the soil–structure interface of foundations.

EXCEPTIONS: The vertical seismic load effect E_v, is permitted to be taken as zero for either of the following conditions:

1. In Eqs. 12.4-1, 12.4-2, 12.4-5, and 12.4-6 where S_{DS} is equal to or less than 0.125.
2. In Eq. 12.4-2 where determining demands on the soil–structure interface of foundations.



12.3.4 Redundancy. A redundancy factor, ρ, shall be assigned to the seismic force-resisting system in each of two orthogonal directions for all structures in accordance with this section.

12.3.4.1 Conditions Where Value of ρ is 1.0. The value of ρ is permitted to equal 1.0 for the following:

1. Structures assigned to Seismic Design Category B or C;
2. Drift calculation and P-delta effects;
3. Design of nonstructural components;
4. Design of nonbuilding structures that are not similar to buildings;
5. Design of collector elements, splices, and their connections for which the seismic load effects, including overstrength of Section 12.4.3, are used;
6. Design of members or connections where the seismic load effects, including overstrength of Section 12.4.3, are required for design;
7. Diaphragm loads determined using Eq. (12.10-1), including the limits imposed by Eqs. (12.10-2) and (12.10-3);
8. Structures with damping systems designed in accordance with Chapter 18; and
9. Design of structural walls for out-of-plane forces, including their anchorage.

12.3.4.2 Redundancy Factor, ρ, for Seismic Design Categories D through F. For structures assigned to Seismic Design Category D and having extreme torsional irregularity as defined in Table 12.3-1, Type 1b, ρ shall equal 1.3. For other structures assigned to Seismic Design Category D and for structures assigned to Seismic Design Categories E or F, ρ shall equal 1.3 unless one of the following two conditions is met, whereby ρ is permitted to be taken as 1.0. A reduction in the value of ρ from 1.3 is not permitted for structures assigned to Seismic Design Category D that have an extreme torsional irregularity (Type 1b). Seismic Design Categories E and F are not also specified because extreme torsional irregularities are prohibited (see Section 12.3.3.1).

a. Each story resisting more than 35% of the base shear in the direction of interest shall comply with Table 12.3-3.
b. Structures are regular in plan at all levels provided that the seismic force-resisting systems consist of at least two bays of seismic force-resisting perimeter framing on each side of the structure in each orthogonal direction at each story resisting more than 35% of the base shear. The number of bays for a shear wall shall be calculated as the length of shear wall divided by the story height or two times the length of shear wall divided by the story height, h_{sx}, for light-frame construction.

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